

Calibration Approaches Summary

CamMap baseline vs. NMC3D depth-balanced calibration variants

GROUND TRUTH

Rotation

180 deg around Y

Translation

[0.0000, 0.2600, -0.7600]

Error Vector

estimate - ground_truth

The report focuses on the primary first-pose optimization result. Final-pose optimization was removed from the comparison because it behaved as a drift-sensitive consistency check rather than the best calibration estimate.

Current / final NMC3D name: [Frame-to-frame equal-width distance-balanced CamMap with 8x8 image-grid selection.](#)

Primary Results Across Datasets

Translation difference is estimate minus ground truth in meters. Smaller component magnitudes mean closer agreement with Gazebo ground truth.

Dataset	Approach	Translation	Difference [dx, dy, dz]	Rot. Err.	Inliers
Ground truth	Ground truth	[0.0000, 0.2600, -0.7600]	[0.0000, 0.0000, 0.0000]	0.000 deg	-
Good recording	CamMap baseline	[0.0354, 0.2543, -0.7506]	[+0.0354, -0.0057, +0.0094]	0.849 deg	3815
Good recording	Current NMC3D: frame-to-frame equal-width distance-balanced CamMap	[0.0460, 0.2529, -0.7615]	[+0.0460, -0.0071, -0.0015]	0.925 deg	1664
Multi-depth recording	CamMap baseline	[-0.0186, 0.2692, -0.7278]	[-0.0186, +0.0092, +0.0322]	0.436 deg	478
Multi-depth recording	Current NMC3D: frame-to-frame equal-width distance-balanced CamMap	[-0.0179, 0.2685, -0.7620]	[-0.0179, +0.0085, -0.0020]	0.640 deg	176

Good Recording: Tested Variants

Dataset: orbcilib-master/good_results_2. Current result: NMC3D/good_results_2/NMC3D_calibration.txt.

Approach	Method	Translation	Difference [dx, dy, dz]	Rot. Err.	Inliers
CamMap baseline	Original CamMap matching and global Sim3 calibration	[0.0354, 0.2543, -0.7506]	[+0.0354, -0.0057, +0.0094]	0.849 deg	3815
Global depth tertiles + 8x8 selected optimization	Global depth-balanced selected matches used for optimization	[0.0354, 0.2543, -0.7506]	[+0.0354, -0.0057, +0.0094]	0.849 deg	3815
Global thresholds scoring, all-depth optimization	Depth-balanced subsets select keyframe pairs; all-depth matches optimize	[0.0294, 0.2523, -0.7475]	[+0.0294, -0.0077, +0.0125]	0.834 deg	3305
Frame-to-frame Euclidean tertiles + 8x8 grid	Each pair is locally balanced by Euclidean distance tertiles	[0.0355, 0.2542, -0.7507]	[+0.0355, -0.0058, +0.0093]	0.850 deg	3815
Current NMC3D: equal-width distance-balanced CamMap	Equal-width Euclidean distance bins plus 8x8 image-grid selection	[0.0460, 0.2529, -0.7615]	[+0.0460, -0.0071, -0.0015]	0.925 deg	1664

Multi-Depth Recording

This dataset has stronger depth variation in the scene, where the current NMC3D balancing is most useful.

Approach	Method	Translation	Difference [dx, dy, dz]	Rot. Err.	Inliers
CamMap baseline	Original CamMap matching and global Sim3 calibration	[-0.0186, 0.2692, -0.7278]	[-0.0186, +0.0092, +0.0322]	0.436 deg	478
Current NMC3D: equal-width distance-balanced CamMap	Equal-width Euclidean distance bins plus 8x8 image-grid selection	[-0.0179, 0.2685, -0.7620]	[-0.0179, +0.0085, -0.0020]	0.640 deg	176

What The Current NMC3D Code Does

- 1 Run normal CamMap detection: BoW, Sim3 initialization, projection matching, and geometric validation.
- 2 For each accepted keyframe pair, keep only matches already found by CamMap.
- 3 For each matched map-point pair, compute average Euclidean camera distance: $0.5 * (||Pc1|| + ||Pc2||)$.
- 4 Split local distances into equal-width near, middle, and far bins.
- 5 Select equal counts from the three bins, limited by the smallest bin.
- 6 Within each bin, use an 8x8 image grid so selected matches are spatially distributed.
- 7 Optimize calibration using the selected frame-to-frame balanced matches.

Depth-Balancing Variants Tested

Variant	Methodology
CamMap baseline	No explicit depth balancing. The optimizer receives the raw accepted CamMap matches.
Global depth tertiles + 8x8 selected optimization	After CamMap matching, sort all accepted matches globally by source-camera z depth, split into near/middle/far tertiles, spread each tertile over an 8x8 image grid, and optimize only selected matches.
Global thresholds + local random uniform selection	Compute global near/far thresholds from source atlas depths, then locally select equal near/middle/far matches per keyframe pair using random/uniform selection. No saved numeric output was present for this variant.
Global thresholds scoring, all-depth optimization	Use balanced subsets only to score/select keyframe pairs, then run optimization with all original all-depth matches.
Frame-to-frame Euclidean tertiles + 8x8 grid	For each accepted keyframe pair, compute average Euclidean camera distance, split into near/middle/far tertiles, distribute over an 8x8 source-image grid, and optimize selected matches.
Current NMC3D equal-width distance-balanced CamMap	Same frame-to-frame idea, but uses equal-width near/middle/far distance bins instead of pure equal-count tertiles.

Final Conclusion

On the good recording, CamMap is already excellent. Current NMC3D makes the z component almost exact, but increases the x offset slightly.

Good recording differences

CamMap: [+0.0354, -0.0057, +0.0094]
Current NMC3D: [+0.0460, -0.0071, -0.0015]

On the multi-depth recording, current NMC3D gives the best translation vector overall, mainly by fixing the depth-axis component.

Multi-depth differences

CamMap: [-0.0186, +0.0092, +0.0322]
Current NMC3D: [-0.0179, +0.0085, -0.0020]

Bottom line: the final version is most useful when the scene contains meaningful depth diversity. It reduces depth-axis bias by preventing the optimizer from being dominated by whichever distance range produced the most raw CamMap matches.